

Tswalu Steel Fabrication

Network Requirements

Functional and Technical Requirements for the Cisco Packet Tracer Implementation

1. Network Objectives

The network must provide a structured, segmented and testable infrastructure for Tswalu Steel Fabrication. The final implementation must support departmental separation, dual-stack IPv4/IPv6 communication, inter-VLAN routing and a dedicated guest wireless network.

2. Functional Requirements

ID	Requirement	Implementation Evidence
FR-01	Separate organisational traffic into five VLANs.	VLANs 10, 20, 30, 40 and 50 configured on SW1.
FR-02	Provide a management network.	VLAN 10 – MANAGEMENT.
FR-03	Provide a production network.	VLAN 20 – PRODUCTION.
FR-04	Provide a finance network.	VLAN 30 – FINANCE.
FR-05	Provide an IT services network.	VLAN 40 – IT_SERVICES.
FR-06	Provide a dedicated guest network.	VLAN 50 – GUEST with AP-Guest/Laptop-Guest.
FR-07	Allow routing between VLANs where permitted.	Router-on-a-stick subinterfaces on R1.
FR-08	Support IPv4 and IPv6.	IPv4 and /64 IPv6 addressing configured on all VLAN gateways.
FR-09	Carry VLAN traffic between SW1 and R1.	802.1Q trunk on SW1 Gi0/1.
FR-10	Provide verification evidence.	Show-command and ping screenshots stored in GitHub.

3. Technical Requirements

The implementation uses one Cisco router and one Cisco switch. SW1 must use access ports for end devices and an 802.1Q trunk toward R1. R1 must use subinterfaces for each VLAN and provide the corresponding IPv4 and IPv6 default gateways.

4. Addressing Requirements

VLAN	IPv4 Network	Gateway	IPv6 Prefix	Gateway
10	172.30.12.64/27	172.30.12.65	2001:DB8:ACAD:10::/64	2001:DB8:ACAD:10::1
20	172.30.12.0/26	172.30.12.1	2001:DB8:ACAD:20::/64	2001:DB8:ACAD:20::1
30	172.30.12.128/28	172.30.12.129	2001:DB8:ACAD:30::/64	2001:DB8:ACAD:30::1
40	172.30.12.144/28	172.30.12.145	2001:DB8:ACAD:40::/64	2001:DB8:ACAD:40::1
50	172.30.12.96/27	172.30.12.97	2001:DB8:ACAD:50::/64	2001:DB8:ACAD:50::1

5. Security Requirements

Departmental and guest traffic must be logically separated using VLANs. Router management must use protected privileged access and authenticated console/VTY access. Guest devices must not be placed directly in internal departmental VLANs.

6. Verification Requirements

The completed project should demonstrate: VLAN membership, trunk operation, IPv4 gateway reachability, IPv6 gateway reachability, inter-VLAN connectivity where required, and guest isolation. The project evidence includes screenshots for these verification activities.

7. Deliverables

The portfolio should contain the Packet Tracer .pkt project, router and switch configuration text files, addressing spreadsheet, topology diagrams, design and requirements documentation, reflection, and verification screenshots.

Source Note

This requirements document records the technical requirements represented by the final implemented Packet Tracer topology and configuration evidence available in the project. It does not claim additional requirements that were not represented in the supplied implementation evidence.